

Bernoulli Mixture Models

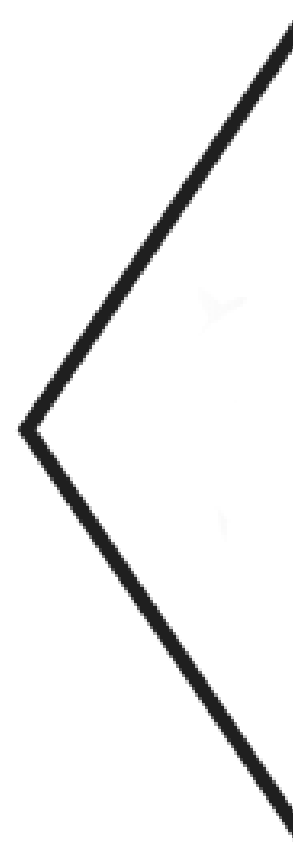
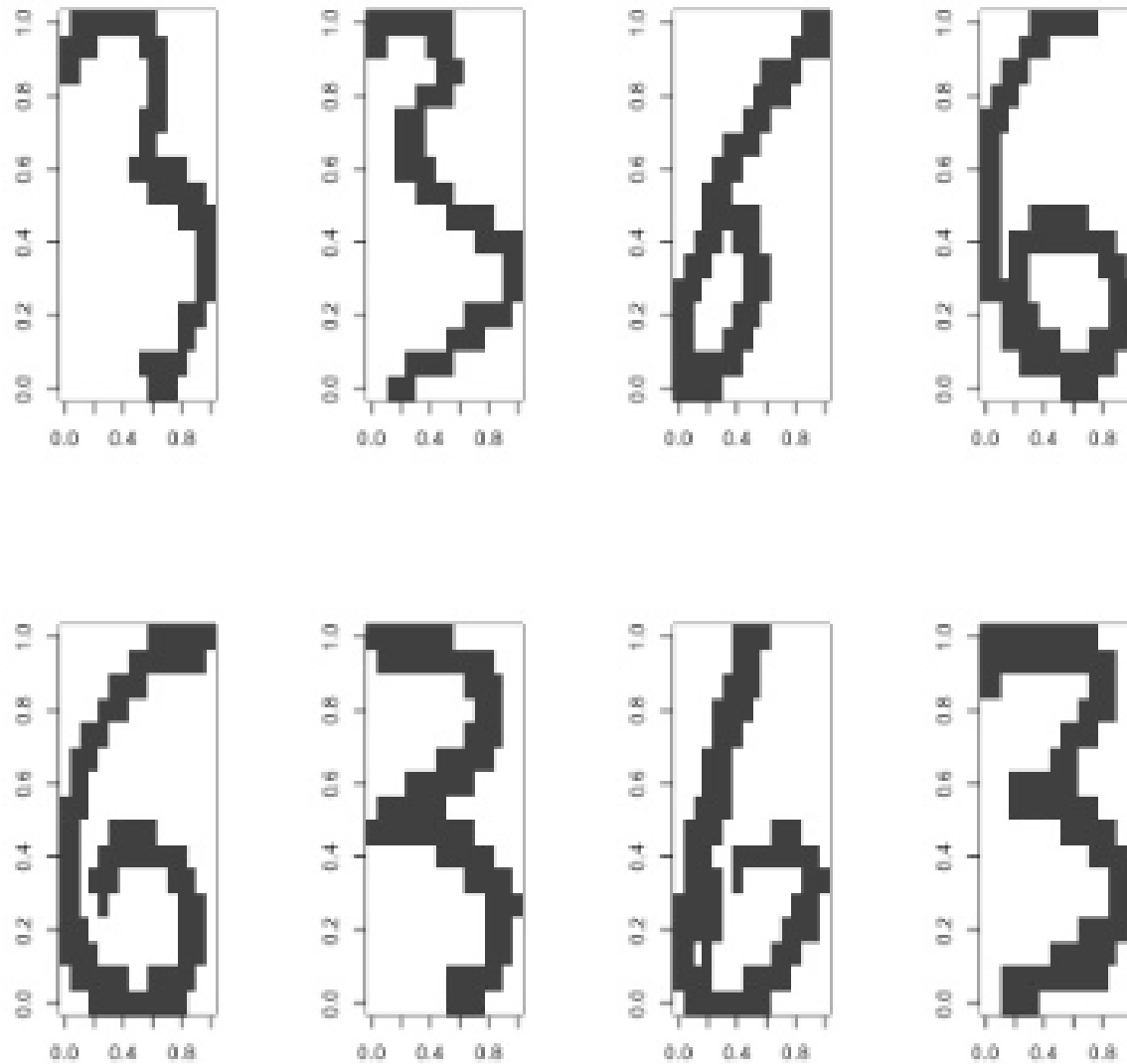
MIXTURE MODELS IN R



Victor Medina

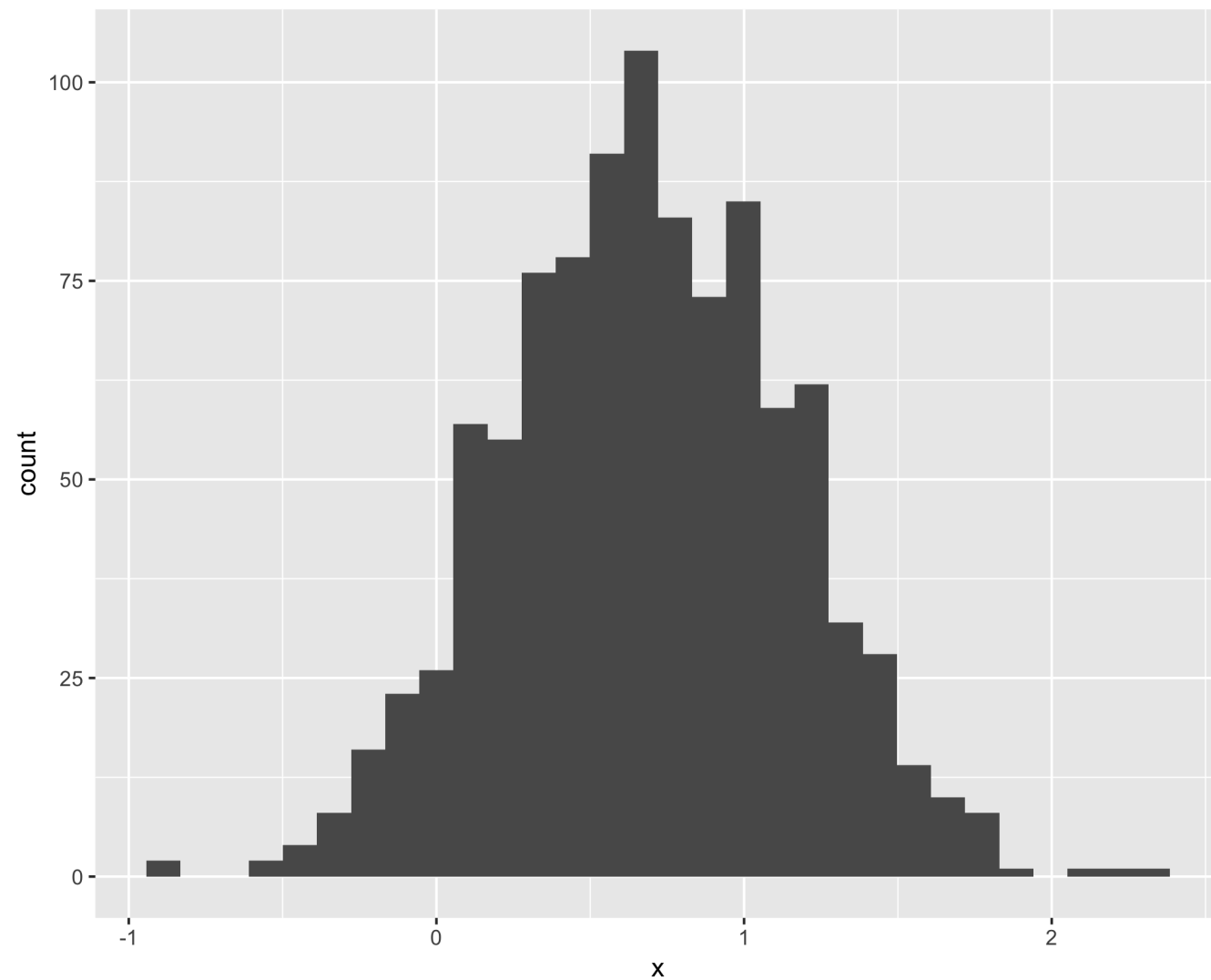
Researcher at The University of
Edinburgh

The handwritten digits dataset

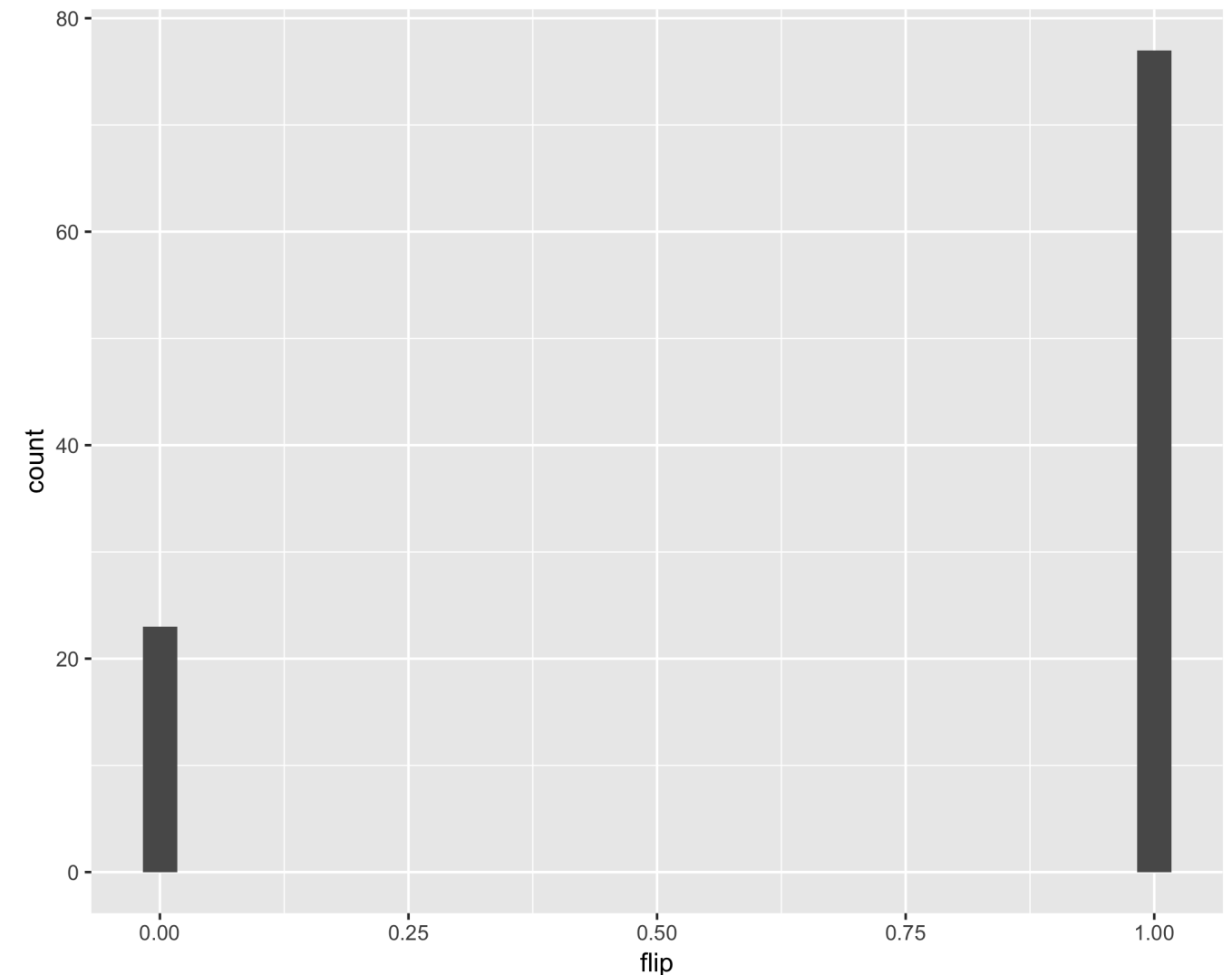


Continuous versus discrete variables

Gaussian distribution



Bernoulli distribution (flipping a coin)



Bernoulli distribution

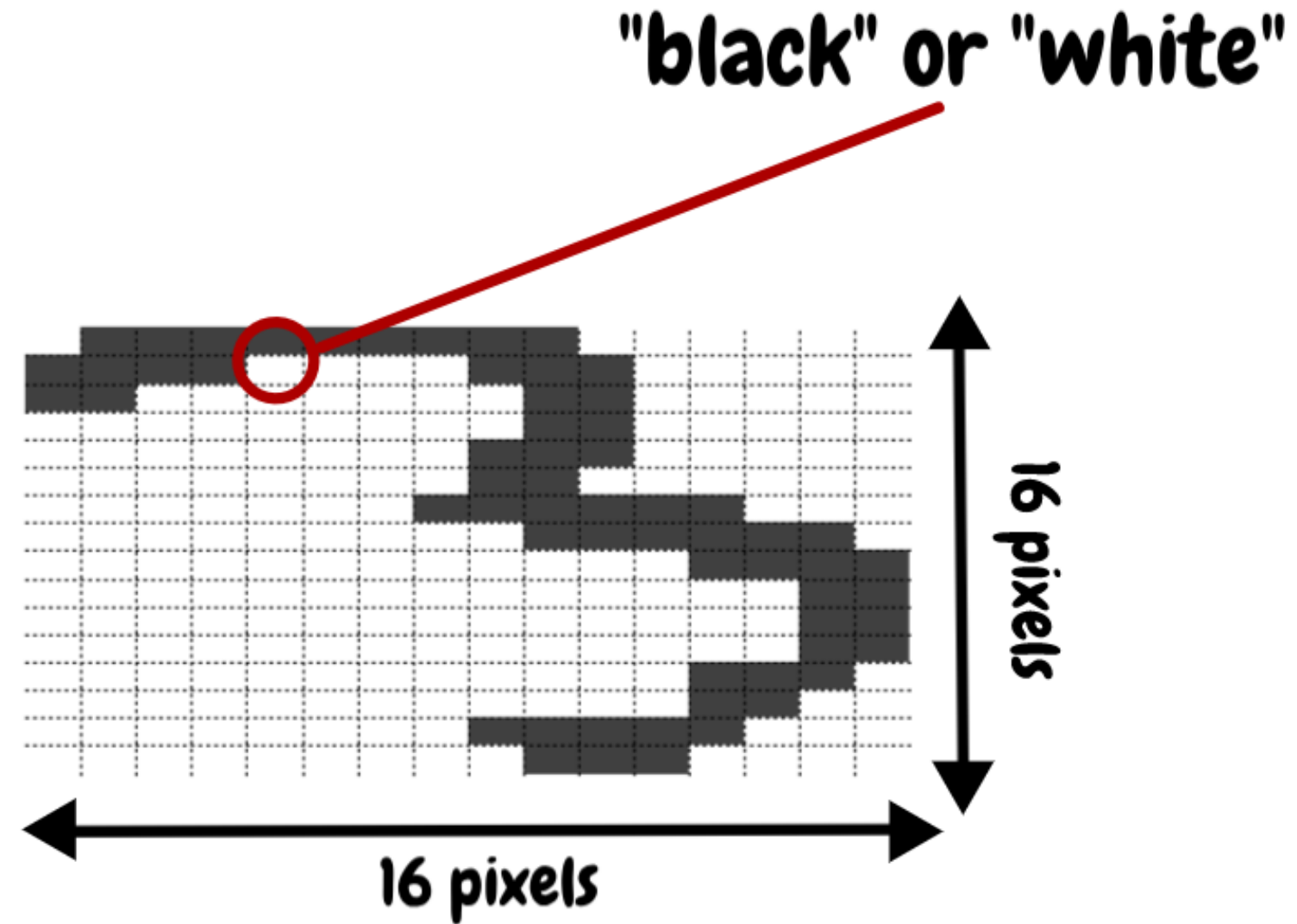
- Two possible outcomes
 - "tails" or "heads"
 - "black" or "white"
- Represented by a probability of "success" $\rightarrow p$
 - $(1 - p)$ = probability for the other option

Sample of Bernoulli distribution

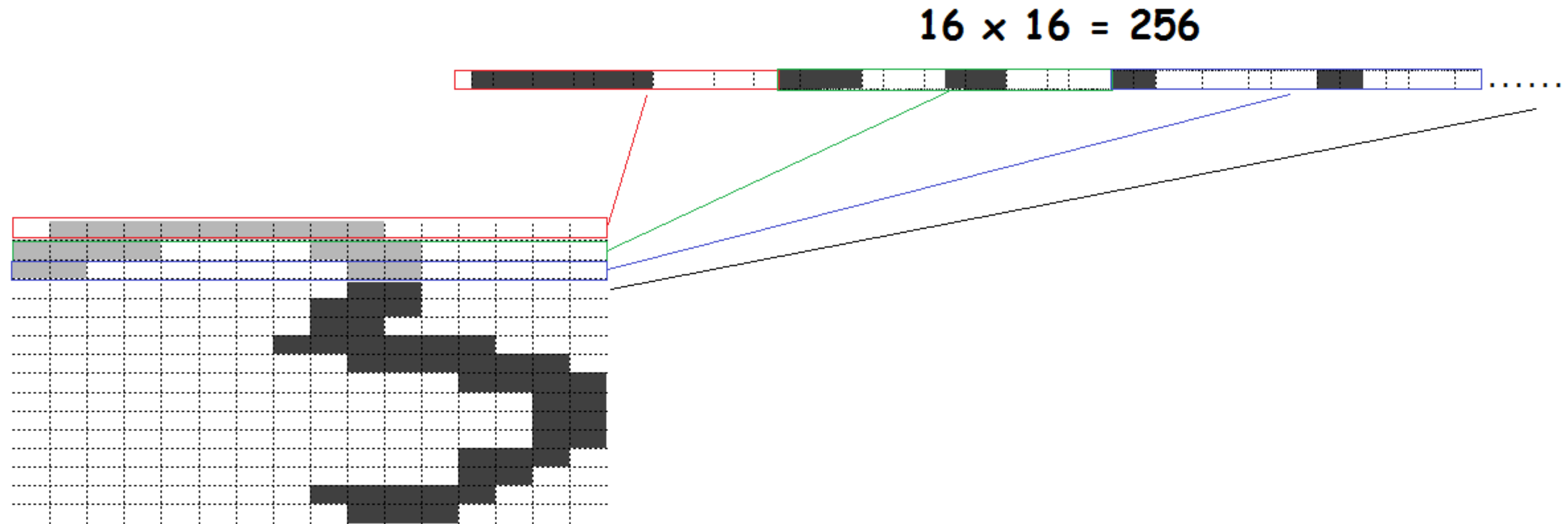
```
p <- 0.7  
bernoulli <- sample(c(0, 1), 100, replace = TRUE, prob = c(1-p, p))  
head(bernoulli)
```

```
1 1 1 0 0 1
```

Binary image as Bernoulli distributions



Binary image as Bernoulli vector




```
p1 <- 0.7; p2 <- 0.5; p3 <- 0.4
```

```
bernoulli_1 <- sample(c(0, 1), 100, replace = TRUE, prob = c(1-p1, p1))
```

```
bernoulli_2 <- sample(c(0, 1), 100, replace = TRUE, prob = c(1-p2, p2))
```

```
bernoulli_3 <- sample(c(0, 1), 100, replace = TRUE, prob = c(1-p3, p3))
```

```
multi_bernoulli <- cbind(bernoulli_1, bernoulli_2, bernoulli_3)
```

```
head(multi_bernoulli, 4)
```

```
      bernoulli_1 bernoulli_2 bernoulli_3
[1,]           1           0           0
[2,]           0           0           0
[3,]           0           0           1
[4,]           1           0           0
```

```
p_vector <- c(p1, p2, p3)
```


Bernoulli mixture models

Handwritten digits dataset:

1. Which is the suitable probability distribution?
 - (multivariate) Bernoulli distribution.
2. How many subpopulations should we consider?
 - Let's try with two. That is two binary vectors of size 256.
3. Which are the parameters and their estimations?
 - Each p for each binary vector. Also the two proportions.

Let's practice

MIXTURE MODELS IN R

Bernoulli Mixture Models with flexmix

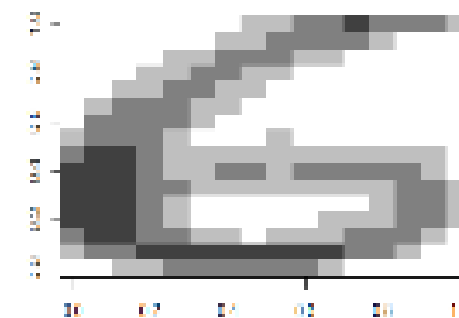
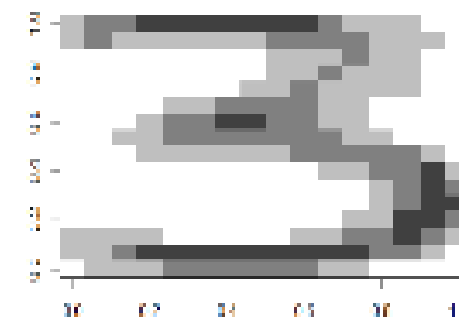
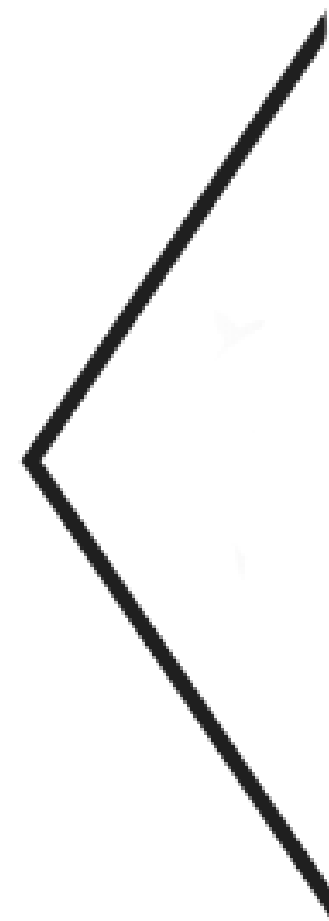
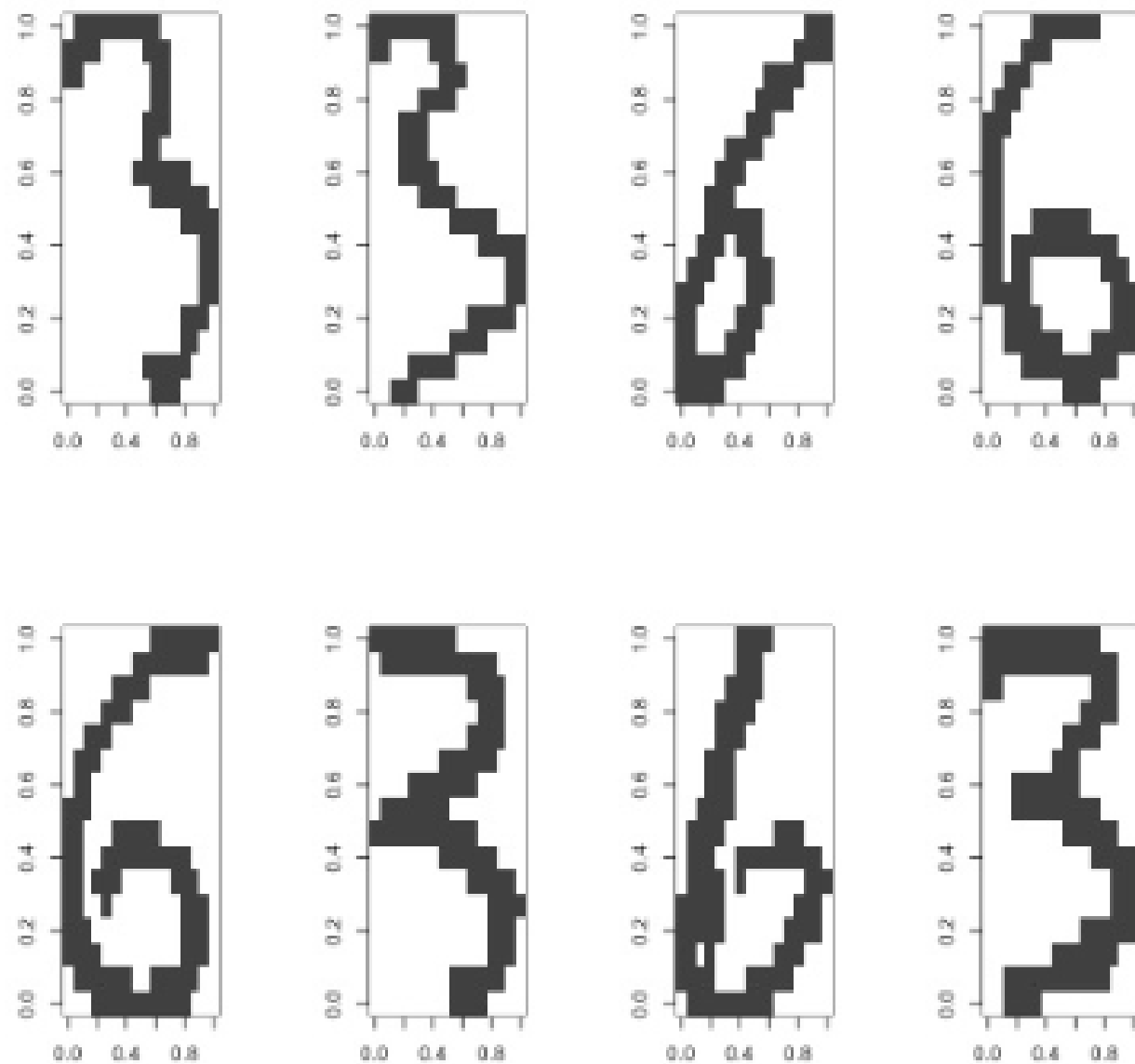
MIXTURE MODELS IN R



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The problem

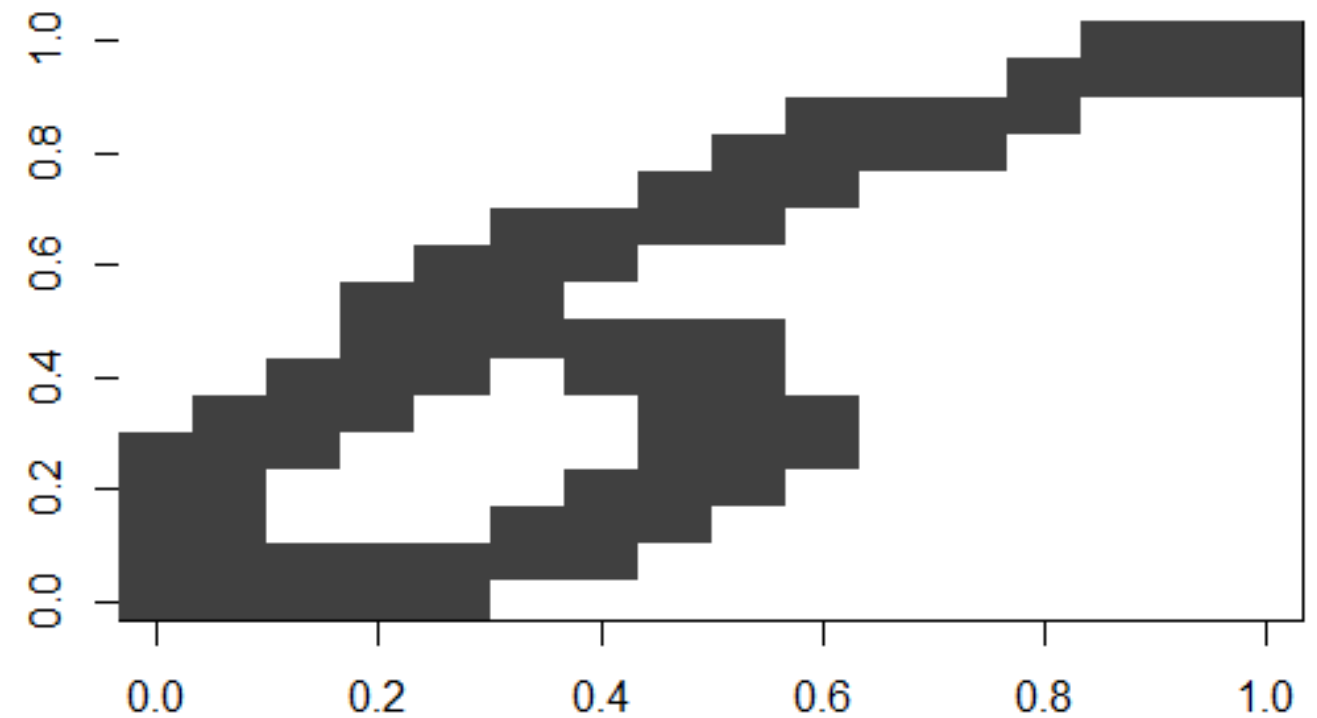


The dataset

```
digits_sample <- as.matrix(digits)
dim(digits_sample)
```

```
320 256
```

```
show_digit(digits_sample[320,])
```



Fit Bernoulli mixture models

```
bernoulli_mix_model <- flexmix(digits_sample~1,  
                                k=2,  
                                model=FLXMCmvbinary(),  
                                control = list(tolerance = 1e-15, iter.max = 1000))
```

- `digits_sample` is a matrix
- `FLXMCmvbinary()` specifies the Bernoulli distribution

The proportions

```
prior(bernoulli_mix_model)
```

```
0.503125 0.496875
```


parameters() function

```
param_comp1 <- parameters(bernoulli_mix_model, component = 1)
param_comp2 <- parameters(bernoulli_mix_model, component = 2)
dim(param_comp1)
```

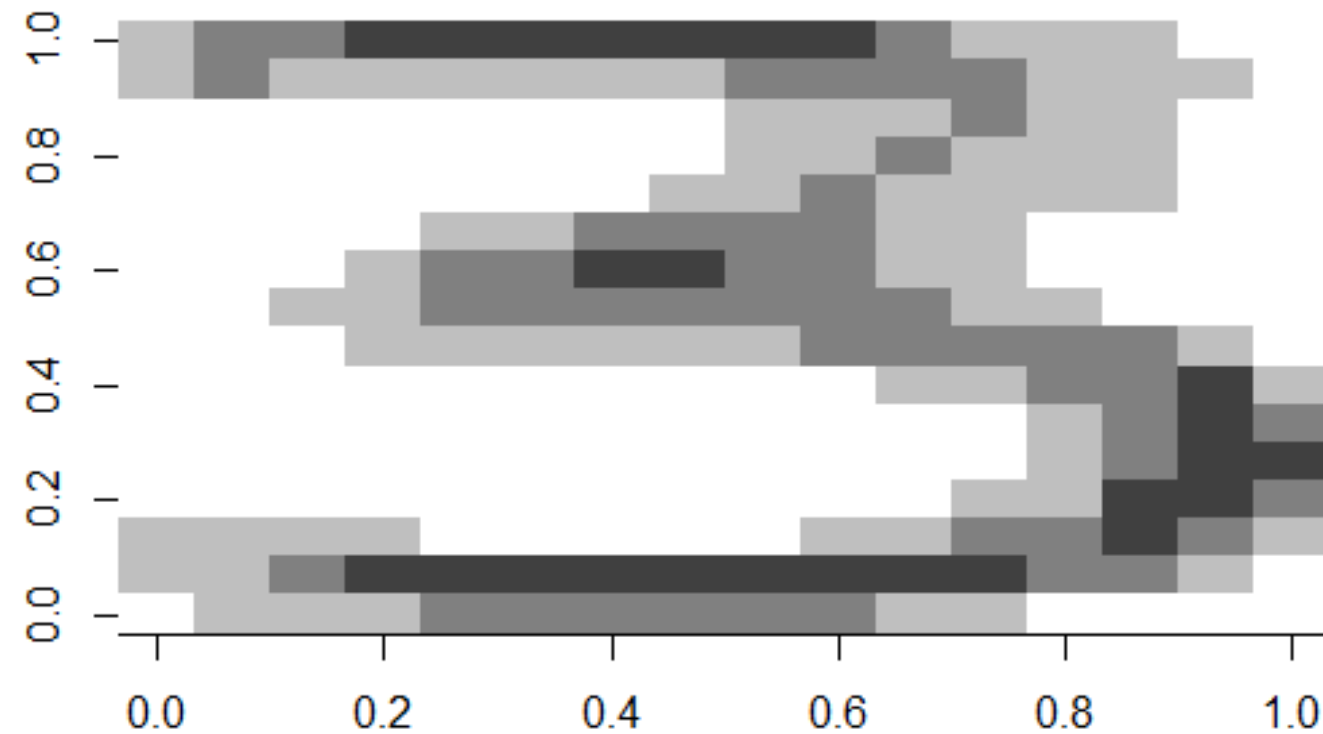
```
256  1
```

```
head(param_comp1, 4)
```

```
      Comp.1
center.V1 0.3291926
center.V2 0.5093168
center.V3 0.6645963
center.V4 0.7639751
```

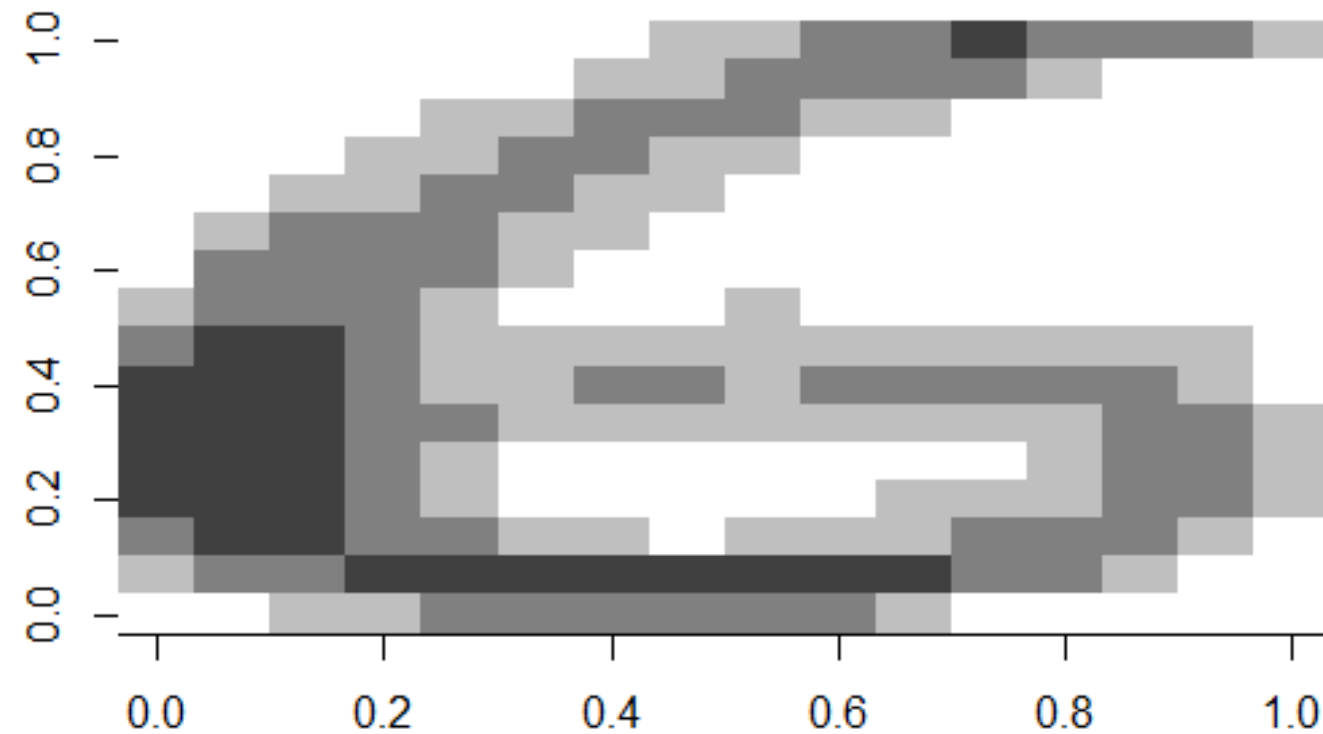

Visualize the component 1

```
show_digit(param_comp1)
```



Visualize the component 2

```
show_digit(param_comp2)
```



Let's practice!
MIXTURE MODELS IN R

Poisson Mixture Models

MIXTURE MODELS IN R



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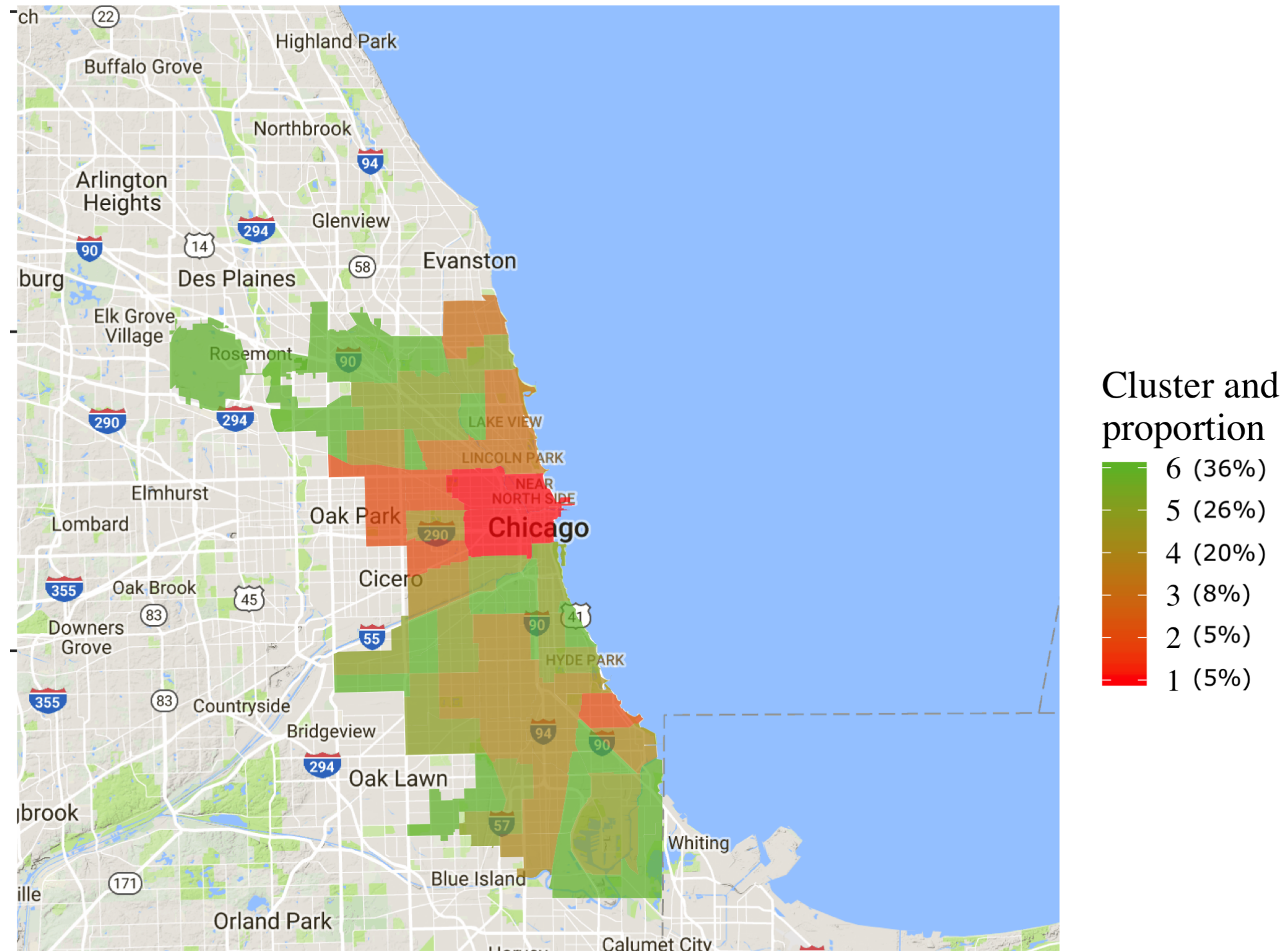
```
# Have a look at the data
```

```
glimpse(crimes)
```

```
Observations: 77
```

```
Variables: 13
```

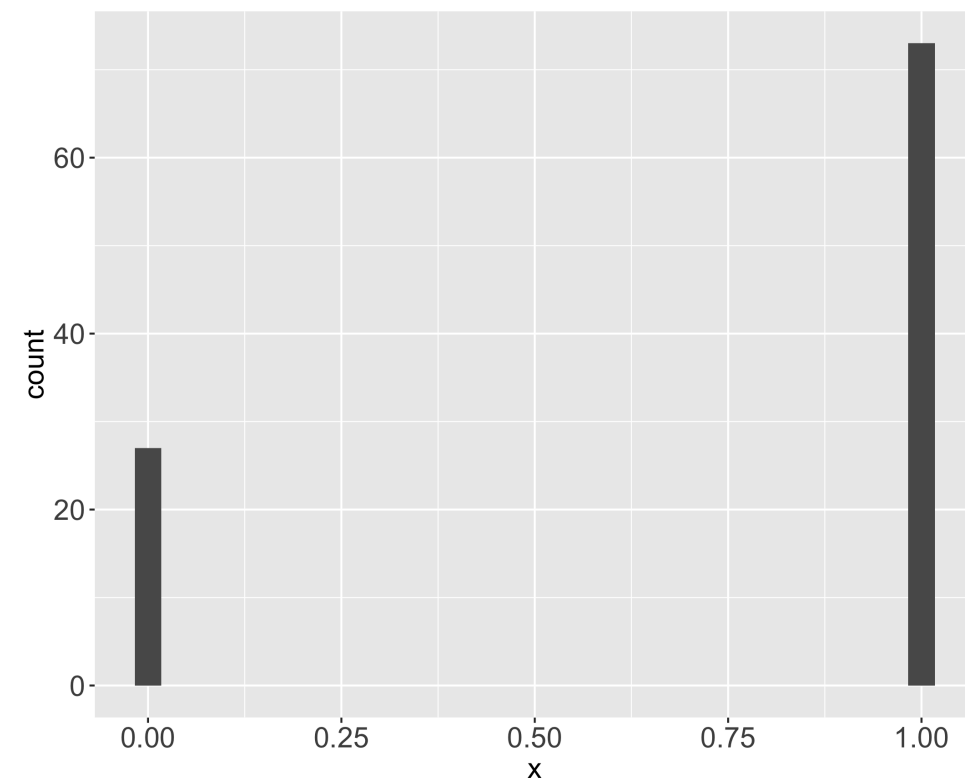
```
$ COMMUNITY      <chr> "ALBANY PARK", "ARCHER HEIGHTS", "...  
$ ASSAULT        <int> 123, 51, 74, 169, 708, 1198, 118, ...  
$ BATTERY        <int> 429, 134, 184, 448, 1681, 3347, 28...  
$ BURGLARY       <int> 147, 92, 55, 194, 339, 517, 76, 14...  
$ `CRIMINAL DAMAGE` <int> 287, 114, 99, 379, 859, 1666, 150,...  
$ `CRIMINAL TRESPASS` <int> 38, 23, 56, 43, 228, 265, 29, 36, ...  
$ `DECEPTIVE PRACTICE` <int> 137, 67, 59, 178, 310, 767, 73, 20...  
$ `MOTOR VEHICLE THEFT` <int> 176, 50, 37, 189, 281, 732, 58, 12...  
$ NARCOTICS      <int> 27, 18, 9, 30, 345, 1456, 15, 22, ...  
$ OTHER          <int> 107, 37, 48, 114, 584, 1261, 76, 8...  
$ `OTHER OFFENSE` <int> 158, 44, 35, 164, 590, 1130, 94, 1...  
$ ROBBERY        <int> 144, 30, 98, 111, 349, 829, 65, 10...  
$ THEFT          <int> 690, 180, 263, 461, 1201, 2137, 23...
```



Comparison of Poisson with Bernoulli

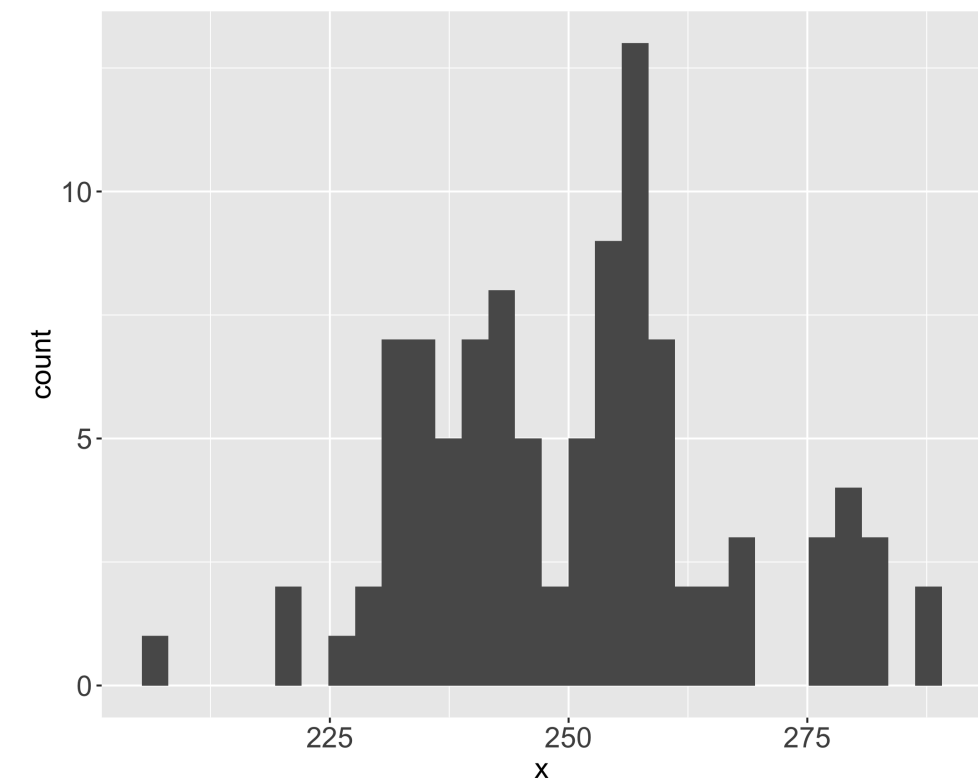
Bernoulli distribution

```
data.frame(x = bernoulli) %>%  
  ggplot(aes(x = x)) + geom_histogram()
```



Poisson distribution

```
data.frame(x = rpois(100, 250)) %>%  
  ggplot(aes(x = x)) + geom_histogram()
```



Poisson distribution

- Number of times an event occurs in an interval of time
- Examples:
 - Number of car accidents in a year
 - Number of emails received in a day
 - Number of robberies in an area of the city for a period of one year

Sample of Poisson distribution

```
lambda_1 <- 100  
poisson_1 <- rpois(n = 100, lambda = lambda_1)  
head(poisson_1)
```

```
98  98  87  77 102  85
```

```
lambda_1 <- 100
lambda_2 <- 200
lambda_3 <- 300

poisson_1 <- rpois(n = 100, lambda = lambda_1)
poisson_2 <- rpois(n = 100, lambda = lambda_2)
poisson_3 <- rpois(n = 100, lambda = lambda_3)

multi_poisson <- cbind(poisson_1, poisson_2, poisson_3)

head(multi_poisson, 4)
```

	poisson_1	poisson_2	poisson_3
[1,]	98	198	296
[2,]	98	213	312
[3,]	87	197	311
[4,]	77	215	299

```
head(crimes)
```

```
# A tibble: 6 x 13
  COMMUNITY      ASSAULT BATTERY BURGLARY `CRIMINAL DAMAGE` `CRIMINAL TRESPASS`
  <chr>          <int>   <int>   <int>          <int>          <int>
1 ALBANY PARK      123     429     147           287             38
2 ARCHER HEIGHTS    51     134      92           114             23
3 ARMOUR SQUARE    74     184      55            99             56
4 ASHBURN          169     448     194           379             43
5 AUBURN GRESHAM   708    1681     339           859            228
6 AUSTIN          1198    3347     517          1666            265
# ... with 7 more variables: `DECEPTIVE PRACTICE` <int>, `MOTOR VEHICLE THEFT` <int>,
#   NARCOTICS <int>, OTHER <int>, `OTHER OFFENSE` <int>, ROBBERY <int>, THEFT <int>
```

Poisson mixture model

1. Which is the suitable probability distribution?
 - (multi) Poisson distribution
2. How many subpopulations should we consider?
 - Let's try from 1 to 15 clusters and pick by BIC.
3. Which are the parameters and their estimations?
 - Each λ for each of the multi Poisson. Also the proportions.

Let's practice!
MIXTURE MODELS IN R

Poisson Mixture Models with flexmix

MIXTURE MODELS IN R



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The problem to solve

1. Which is the suitable probability distribution?
 - (multi) Poisson distribution
2. How many subpopulations should we consider?
 - Let's try from 1 to 15 clusters and pick by BIC.
3. Which are the parameters and their estimations?
 - Each λ for each of the multi Poisson. Also the proportions.


```
crimes_matrix <- as.matrix(crimes[, -1])
```

```
poisson_mix_model <- stepFlexmix(crimes_matrix ~ 1,  
                                k = 1:15,  
                                nrep = 5,  
                                model = FLXMCmvpois(),  
                                control = list(tolerance = 1e-15, iter = 1000))
```

- Use `stepFlexmix` instead of `flexmix` function.
- `k` is now a range of values.
- `nrep` is the number of repetitions the EM algorithm runs for each `k` value.
- The Poisson distribution is `FLXMCmvpois`

Pick the best model

```
best_fit <- getModel(poisson_mix_model, which = "BIC")
```

- Other statistical criteria implemented in `flexmix` are the AIC and ICL.

The proportions

```
prior(best_fit)
```

```
0.07792208 0.05194805 0.19480519 0.27272727 0.20779224 0.19480517
```

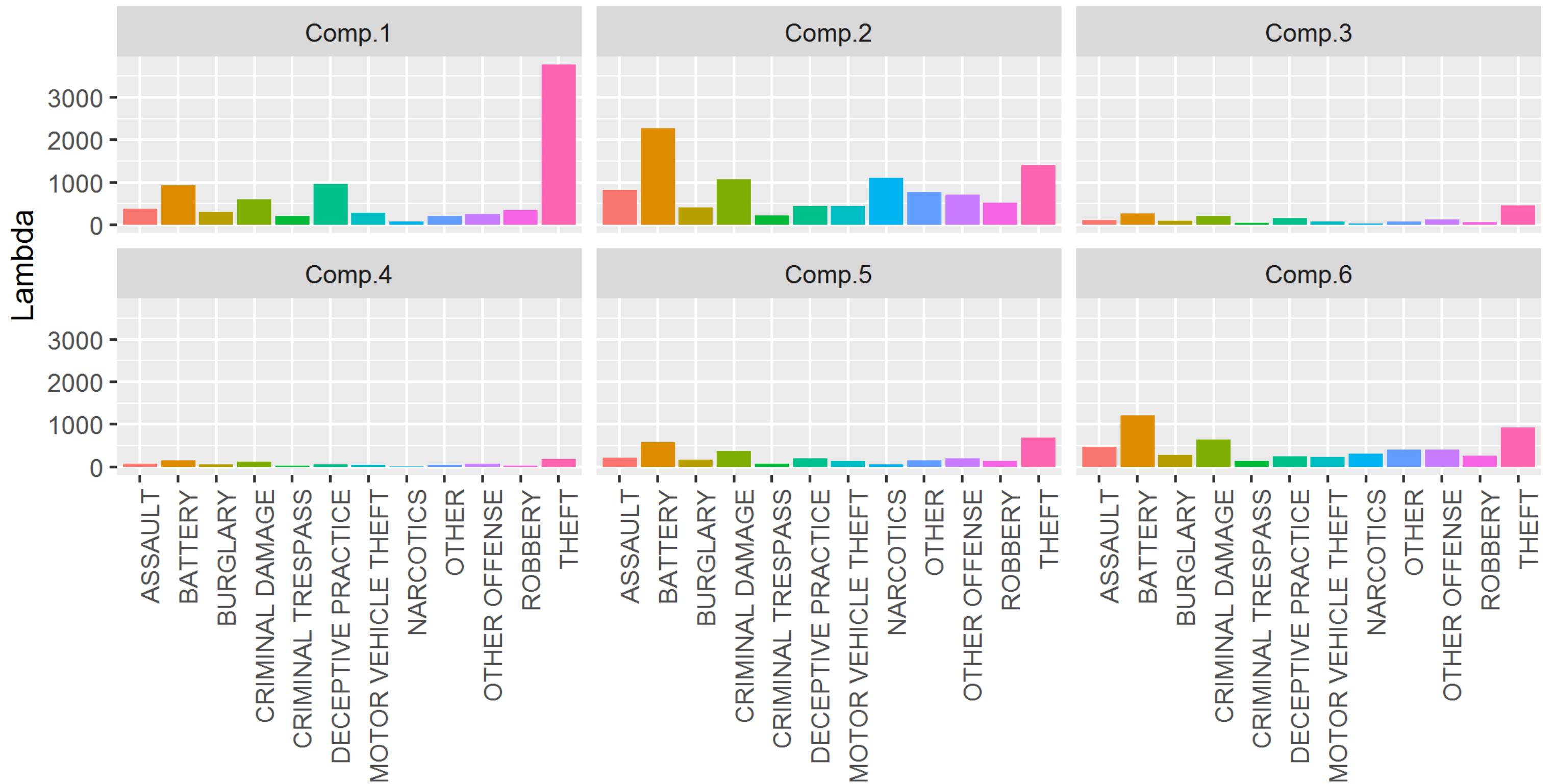
```
param_pmm <- data.frame(parameters(best_fit))
```

```
param_pmm <- param_pmm %>% mutate(Type = colnames(crimes_matrix))  
head(param_pmm)
```

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Type
1	380.3333	821.75	112.26667	67.57143	216.9375	475.3334	ASSAULT
2	929.5000	2271.50	268.13333	153.14286	574.7500	1204.8667	BATTERY
3	303.8333	418.00	98.60000	52.04762	174.9375	272.9333	BURGLARY
4	601.3333	1074.50	199.66666	116.90476	370.9375	648.6667	CRIMINAL DAMAGE
5	210.5000	223.75	49.73333	25.00000	81.0625	139.0000	CRIMINAL TRESPASS
6	973.1667	438.00	158.80000	61.95238	196.7500	241.4666	DECEPTIVE PRACTICE

Visualize the clusters

```
param_pmm %>%  
  pivot_longer(-Type, names_to = "Components", values_to = "Lambda") %>%  
  ggplot(aes(x = Type, y = Lambda, fill = Type)) +  
  geom_bar(stat = "identity") +  
  facet_wrap(~ Components) +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1),  
        legend.position = "none")
```

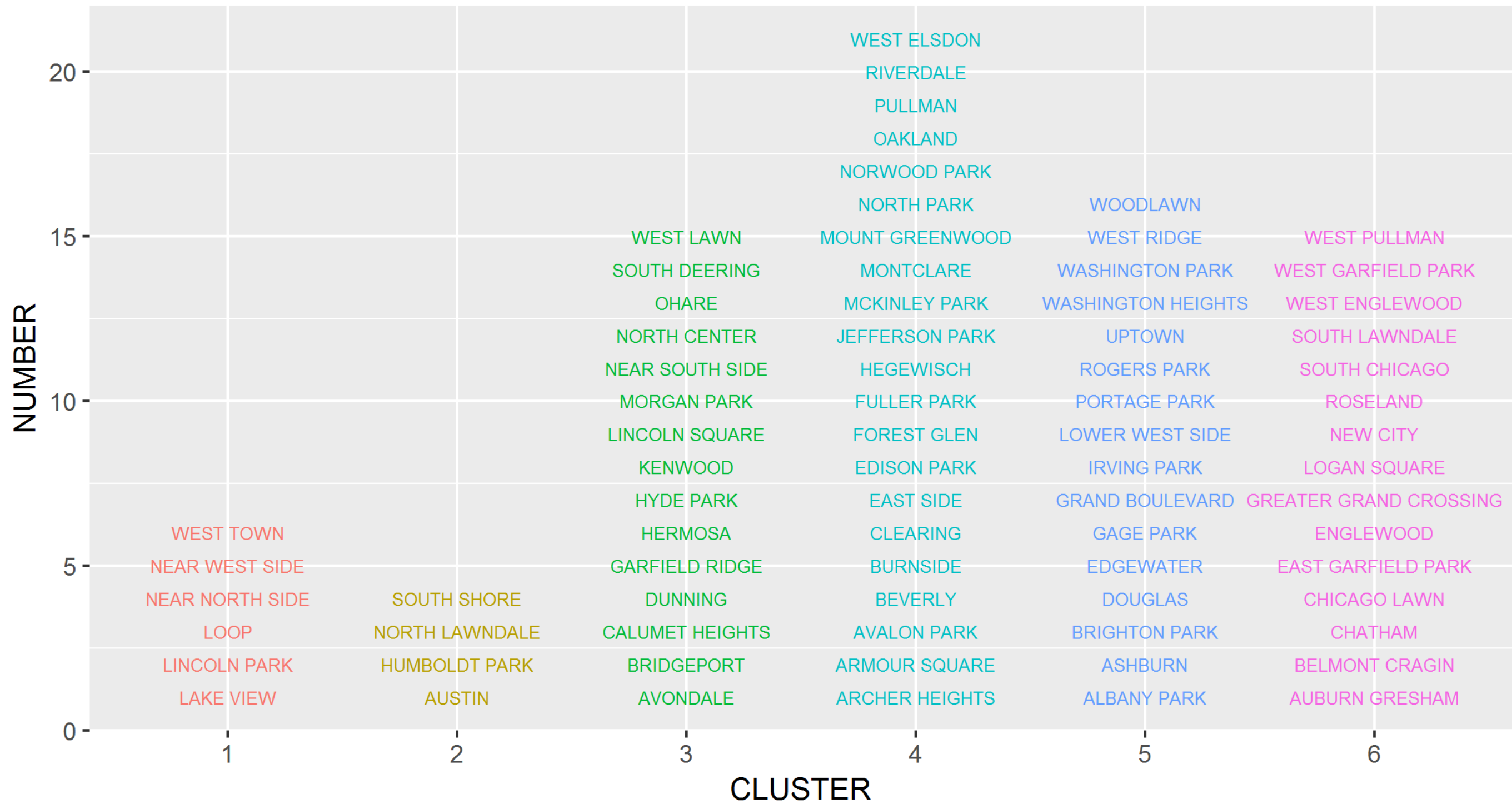


Assign cluster to each community

```
crimes_c <- crimes %>%  
  mutate(CLUSTER = factor(clusters(best_fit)))
```

Visualize the clusters with their communities

```
crimes_c %>%  
  group_by(CLUSTER) %>%  
  mutate(NUMBER = row_number()) %>%  
  ggplot(aes(x = CLUSTER, y = NUMBER, col = CLUSTER)) +  
  geom_text(aes(label = COMMUNITY), size = 2.3)+  
  theme(legend.position="none")
```

Let's practice!
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